Data Analysis Programming
Semester 2026-II
Team Project Assignment
Natural Hazard Monitoring Systems

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Team Project Assignment

All teams build a system with the same general objective: a multi-source data ingestion and analysis system that integrates historical and continuously updated information about a natural hazard, detects new events or changes, and produces analytical indicators about the current situation. The technical contract, data model, and rubric are identical for every team (see the *Course Project* document); only the hazard, sources, and indicator differ.

Important: the 26 enrolled students were split into groups of three following the order of the official course roster, which yields eight teams of three students and one team of two (Team 9). Each team is assigned one hazard system, in order. The professor confirms the assignment at the beginning of Workshop #1. Sources listed are *suggestions to be validated*: availability, access keys, rate limits, and licenses must be verified by each team in Workshop #1, and an equivalent alternative source may be approved by the professor.

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Team Project Distribution

Team	Hazard System	Hazard Indicator / Output	Students
1	Earthquake Activity Monitoring	Event frequency, magnitude and depth distributions, unusual-activity flag, cross-source matching	20211020097 Juan Córdoba 20212020007 Angelo Ibáñez 20212020045 Mauricio Sánchez
2	Flood Monitoring and Risk	Flood Risk Index (LOW / MEDIUM / HIGH or numeric)	20212020080 Oscar Durán 20221020146 Laura Pardo 20222020023 Nicolás Romero
3	Wildfire Monitoring	Anomalous fire activity per region (today, last 7 days, historical average)	20222020059 César Ospina 20222020093 Santiago Alfonso 20222020104 Adrián Espinosa
4	Tropical Cyclone Monitoring	Comparison of the current storm with similar historical storms	20222020108 Omar Peña 20222020134 Rosa López 20222020139 Brayan Ramírez
5	Extreme Weather Monitoring	NORMAL / UNUSUAL / EXTREME from historical percentiles	20222020152 Andrés Mendoza 20222020167 David Martínez 20231020002 Juan Escallón
6	Volcanic Activity Monitoring	Volcano Activity Indicator	20231020006 Juan Rangel 20231020055 Jhonatan Vergara 20231020064 Julián Pulido
7	Landslide Monitoring and Risk	Landslide Risk Indicator per region	20231020072 Samuel Leal 20231020100 Cristian Gámez 20231020116 Sara Martínez
8	Tsunami Monitoring	Potential tsunami relevance of significant earthquakes	20231020133 Lina Moreno 20231020200 Jaider Ávila 20231020205 Gabriela Martínez
9	Multi-Hazard Weather Impact Monitoring	Compound Hazard Index per region	20232020341 Pablo Garzón 20232020356 Kevin Rincón

Geographic Scope and Source Coverage

Before choosing a scope, check *where* each source has data. Several of the richest open sources are national services of the United States and contain no data for Colombia or other countries:

- **United States only:** NOAA Storm Events Database, National Weather Service (NWS) API (observations, forecasts, and alerts), NOAA GHCN-Daily stations accessed through NCEI (mostly US, with sparse global stations), USGS Landslide Inventories, and USGS volcano alert levels (HANS).
- **Regional:** National Hurricane Center products cover only the Atlantic and the Eastern/Central Pacific basins; the tsunami.gov warning centers publish messages for the Pacific, Alaska, US, and Caribbean areas.
- **Global:** USGS earthquake catalog and feeds, EMSC, GDACS, NASA FIRMS, IBTrACS, Smithsonian GVP, NOAA/NCEI historical tsunami database, NASA Global Landslide Catalog, and Open-Meteo.

Teams 2, 5, and 9 therefore work on a **US region** (for example, one or several states), because their required sources are US services. Teams 1, 3, 4, 6, 7, and 8 may choose a global scope, a region, or **Colombia**. Colombian sources (Servicio Geológico Colombiano, IDEAM, UNGRD) are welcome as additional sources, but their programmatic access must be demonstrated in Workshop #1 before they are adopted.

Detailed Hazard Descriptions and Suggested Sources

All sources below were checked on 3 November 2026. They are a starting point: every team must validate access, rate limits, and licenses in Workshop #1. Teams must use the official documentation linked here, not third-party wrappers.

1. **Earthquake Activity Monitoring (Team 1):** Monitor seismic activity by integrating observations from different seismic agencies and identifying new, significant, or unusual events.
   - *Historical 1:* USGS FDSN Event Web Service (`https://earthquake.usgs.gov/fdsnws/event/1/`), queryable by time, magnitude, and area; CSV, GeoJSON, QuakeML. At most 20,000 events per request, so the bulk load must be partitioned by time window.
   - *Historical 2:* EMSC Seismic Portal FDSN service (`https://www.seismicportal.eu/fdsn-wsevent.html`), an independent catalog with its own identifiers, from 1998. Optionally, the Servicio Geológico Colombiano catalog (`https://www.sgc.gov.co/catalogo`) for a Colombian scope.
   - *Incremental:* USGS real-time GeoJSON summary feeds (`https://earthquake.usgs.gov/earthquakes/feed/v1.0/geojson.php`; rolling windows of the past hour, day, 7 days, and 30 days, recommended by USGS for applications) and GDACS events API (`https://www.gdacs.org/gdacsapi/`, type `EQ`, significant events only). EMSC also offers a WebSocket feed of new and updated events.

- *Analytics:* event frequency, magnitude and depth distributions, geographic concentration, unusual activity, cross-source matching.
- *Main challenge:* deciding whether records from different catalogs describe the same earthquake (origin time, epicenter, and magnitude differ slightly, and magnitude types differ).

2. **Flood Monitoring and Risk (Team 2) — US scope:** Combine historical flooding, rainfall, and current environmental observations.
   - *Historical 1:* NOAA Storm Events Database (`https://www.ncei.noaa.gov/stormevents/`), yearly gzipped CSV files from 1950 (`https://www.ncei.noaa.gov/pub/data/swdi/stormevents/csvfiles/`); use the event types Flood, Flash Flood, and Heavy Rain. Files are re-compiled when NOAA revises them (the `_cYYYYMMDD` suffix in the file name changes), and recent months appear with a delay of several months.
   - *Historical 2:* daily precipitation from GHCN-Daily through the NCEI Access Data Service (`https://www.ncei.noaa.gov/access/services/data/v1`, dataset `daily-summaries`, JSON or CSV).
   - *Incremental:* NWS API (`https://www.weather.gov/documentation/services-web-api`) for latest station observations (`/stations/{id}/observations`) and for active alerts (`/alerts/active`), such as Flood and Flash Flood warnings. The API requires a `User-Agent` header identifying the application and a contact.
   - *Analytics:* current rainfall and accumulation, historical rainfall distribution, historical flood frequency, current alerts → Flood Risk Index (LOW / MEDIUM / HIGH or numeric).
   - *Main challenge:* combining temporal rainfall information with geographic flood events (Storm Events locates many events by county or forecast zone, not always by coordinates).

3. **Wildfire Monitoring (Team 3):** Determine whether current fire activity is unusual relative to historical patterns and weather conditions.
   - *Historical 1:* NASA FIRMS active-fire archive (`https://firms.modaps.eosdis.nasa.gov/api/`): MODIS from 2000 and VIIRS (Suomi NPP from 2012, NOAA-20 from 2018). The area API returns CSV for 1–5 days per request, which forces a partitioned bulk load. A free `MAP_KEY` is required, with a limit of 5,000 transactions per 10 minutes.
   - *Historical 2:* historical weather (temperature, humidity, precipitation, wind) from the Open-Meteo Historical Weather API (`https://open-meteo.com/en/docs/historical-weather-api`, ERA5 reanalysis, global, no key).
   - *Incremental:* FIRMS near-real-time detections from **VIIRS NOAA-20 and NOAA-21** (`VIIRS_NOAA20_NRT`, `VIIRS_NOAA21_NRT`), usually available about three hours after the satellite overpass, and current weather from the Open-Meteo Forecast API (`https://open-meteo.com/en/docs`) or NWS for a US scope.

- *Important (2026 sensor changes):* Suomi NPP stopped producing new data on 2 November 2026; its archive remains available, but it is no longer an incremental source. Terra MODIS has been in standby since March 2026, and Aqua MODIS is being retired. The historical baseline therefore mixes sensors with different detection capabilities. Teams must build the baseline per sensor, or harmonize it with an overlapping period, and must document the effect.
- *Analytics:* fires today, last 7 days, historical average per region, and weather context → areas with anomalous activity.
- *Main challenge:* geospatial and temporal aggregation of satellite detections, including grouping detections into fire events.

4. **Tropical Cyclone Monitoring (Team 4):** Track cyclones by integrating historical trajectories, current observations, forecasts, and impacts.

   - *Historical 1:* IBTrACS v04 (`https://www.ncei.noaa.gov/products/international-best-track-archive`), the global best-track archive, in CSV, shapefile, and netCDF, by basin or for all basins.
   - *Historical 2:* NOAA Storm Events (Tropical Storm, Hurricane, and Storm Surge event types) for impacts in the US, or GDACS past tropical-cyclone events for a global scope.
   - *Incremental:* IBTrACS `ACTIVE` subset (storms active in the last 7 days, updated frequently; check the `Last-Modified` header), NHC `CurrentStorms.json` (`https://www.nhc.noaa.gov/CurrentStorms.json`) with its GIS forecast tracks and cones (`https://www.nhc.noaa.gov/gis/`), and GDACS (type `TC`, global).
   - *Important (seasonality):* the Atlantic and Eastern Pacific seasons end on 30 November. To observe live storms during the semester, the team must use a **global** scope, because Southern Hemisphere and Western Pacific storms are covered by IBTrACS and GDACS but not by NHC. An empty feed is a valid response and must be handled. Recorded feeds from past storms must be kept as test fixtures.
   - *Analytics:* current position, speed, direction, intensity, forecast path, and historical similar storms in the same region.
   - *Main challenge:* track processing and trajectory comparison.

5. **Extreme Weather Monitoring (Team 5) — US scope:** Detect unusual extreme heat, cold, heavy precipitation, severe wind, hail, tornadoes, and winter storms.

   - *Historical 1:* NOAA Storm Events Database (bulk CSV, 1950 onward, many event types).
   - *Historical 2:* GHCN-Daily observations (`TMAX`, `TMIN`, `PRCP`, `SNOW`, and others) through the NCEI Access Data Service.
   - *Incremental:* NWS alerts API (watches, warnings, advisories; GeoJSON and CAP) and NWS latest station observations.

- *Analytics:* current measurement versus historical mean, percentile, and extreme for the same station and season → NORMAL / UNUSUAL / EXTREME.
- *Main challenge:* anomaly detection using historical distributions (seasonality and station changes).

6. **Volcanic Activity Monitoring (Team 6):** Integrate historical eruptions, volcano characteristics, current activity reports, and nearby seismicity.

   - *Historical 1:* Smithsonian Global Volcanism Program, *Volcanoes of the World* database (more than 1,200 Holocene volcanoes and over 11,000 Holocene eruptions). It is available through the GVP WFS web service (`https://volcano.si.edu/database/webservices.cfm`), with layers for Holocene volcanoes and Holocene eruptions in GeoJSON, CSV, or shapefile. Cite the database version used.
   - *Historical 2:* the USGS earthquake catalog (FDSN) queried around each volcano's coordinates.
   - *Incremental:* Smithsonian/USGS Weekly Volcanic Activity Report (`https://volcano.si.edu/reports_weekly.cfm`; RSS and CAP feeds; **weekly**, so a slow incremental source) and the USGS real-time earthquake feeds. Optional additions: GDACS (type `VO`) and, for US volcanoes, the USGS HANS API (`https://volcanoes.usgs.gov/hans-public/api/volcano/`) with current alert levels and aviation color codes. For Colombia, the SGC volcano observatories publish activity bulletins (see the Workshop #1 rule for national sources).
   - *Analytics:* historical activity, recent seismicity, current alert level, and eruption history → Activity Indicator.
   - *Main challenge:* integrating events with static geographic entities (earthquake → distance from volcano → activity window → indicator), and linking report text to the volcano number (`vnum`).

7. **Landslide Monitoring and Risk (Team 7):** Identify areas where current rainfall and conditions resemble those associated with historical landslides.

   - *Historical 1 (inventory):* NASA Global Landslide Catalog / COOLR (`https://landslides.nasa.gov`; global, CSV, shapefile, or geodatabase; rainfall-triggered events). Alternatives: USGS *Landslide Inventories across the United States* ver. 3.0 (`https://www.sciencebase.gov/catalog/item/671eef1fd34ed0f827ea9f12`) for a US scope, or SGC SIMMA for Colombia.
   - *Historical 2 (rainfall):* Open-Meteo Historical Weather API (daily and hourly precipitation, global) or GHCN-Daily for a US scope.
   - *Incremental:* current and recent precipitation from the Open-Meteo Forecast API (past days, current, and forecast; global, no key). As a second incremental source, use one of the following: the NASA LHASA global landslide nowcast (`https://maps.nccs.nasa.gov/download/landslides/nowcast`; daily raster, published on a best-effort basis), the ReliefWeb API v2 disasters endpoint

(`https://apidoc.reliefweb.int/`; requires a **pre-approved** `appname`, so request it in the first week), or a national warning feed. **Note:** GDACS does not publish landslide events.

- *Analytics:* rainfall over 24 hours and 7 days, historical rainfall, landslide density, terrain characteristics, and current alerts → Landslide Risk Indicator.
- *Main challenge:* spatial joins between environmental measurements and historical events (inventory locations have stated accuracy radii).

8. **Tsunami Monitoring (Team 8):** Correlate earthquakes, historical tsunami events, and official alerts.
   - *Historical 1:* NOAA/NCEI Global Historical Tsunami Database through the HazEL REST API (`https://www.ngdc.noaa.gov/hazel/view/swagger`; JSON), more than 2,400 events. The related significant-earthquake and run-up tables are available through the same API.
   - *Historical 2:* USGS earthquake catalog (FDSN), for example events of M≥6.
   - *Incremental:* USGS real-time earthquake feeds (the `tsunami` property flags events for which a tsunami bulletin may exist; it does *not* mean a tsunami occurred) and the tsunami.gov warning-center Atom feeds (`https://www.tsunami.gov/events/xml/PAAQAtom.xml` and `https://www.tsunami.gov/events/xml/PHEBAtom.xml`), each entry linking to a CAP message. Most messages are information statements, so absence of a threat is the normal case.
   - *Analytics:* magnitude, depth, location, distance to coast, historical tsunami association, and official alert → potential tsunami relevance.
   - *Main challenge:* correlating earthquakes with spatially and temporally related tsunami information. **Teams must not claim to predict tsunamis**; the goal is event correlation and monitoring.
9. **Multi-Hazard Weather Impact Monitoring (Team 9) — US scope:** Identify areas with unusually high combined weather-hazard activity (heavy rain, strong wind, flood warnings, extreme temperature).
   - *Historical 1:* NOAA Storm Events (several event types for the same regions).
   - *Historical 2:* GHCN-Daily observations through the NCEI Access Data Service.
   - *Incremental:* NWS alerts API and NWS latest station observations.
   - *Analytics:* per-region combination of hazard components → Compound Hazard Index.
   - *Main challenge:* temporal and spatial correlation of multiple event types (alerts are issued for zones and counties, observations for stations).

Access Checklist for Workshop #1

- **Keys and registration:** NASA FIRMS `MAP_KEY` (free, immediate); ReliefWeb `appname` (pre-approval, may take days); NWS needs no key, but a descriptive `User-Agent`; NCEI Access Data Service, USGS, EMSC, GDACS, GVP, HazEL, and Open-Meteo need no key.

- **Limits and terms:** USGS FDSN returns at most 20,000 events per query; FIRMS allows 5,000 transactions per 10 minutes; Open-Meteo's free tier is for non-commercial use (about 10,000 calls per day) and requires attribution (CC BY 4.0). Cache all downloads.
- **Licenses and citation:** record each source's attribution requirement (e.g., GVP database version and DOI, NASA catalog citations) in `config/sources.yaml` and in the report.
- **Volatility:** sources change (satellites end, APIs move to new versions, feeds are renamed). Each team must keep a short source-status log and report any change to the professor as soon as it is detected.

Common Technical Contract

- **Historical ingestion:** at least two sources of different formats or APIs, several years of data, initial bulk load via `python ingest.py -mode historical`.
- **Incremental ingestion:** at least two periodically updated sources, watermark-based polling via `python ingest.py -mode incremental`, new/changed record detection, duplicate handling, incremental storage.
- **Processing:** schema normalization, validation, missing-value handling, duplicate detection, event identification, source integration, temporal analysis, geographic analysis, historical-versus-current comparison.
- **Data model:** `source`, `ingestion_run`, `ingestion_state`, `event` (id, source, type, timestamp, latitude, longitude, severity, raw reference), `event_group`, `rejected_record`, plus domain tables (e.g., `earthquake`, `fire_detection`).
- **Infrastructure:** Docker Compose (database, ingestion application, orchestrator, dashboard), configuration via environment variables and `config/sources.yaml`.
- **Dashboard:** Plotly Dash monitoring application at `localhost:8050`.

Workshop and Delivery Schedule

- **Workshop #1:** Hazard definition, validation of four sources, profiling, canonical mapping, ingestion contract draft, and project definition.
- **Workshop #2:** Architecture, Docker stack, database schema, and historical ingestion with validation and quarantine.
- **Workshop #3:** Incremental ingestion, watermarks, scheduling, change detection, and event identification across sources.
- **Workshop #4:** Temporal and geographic analysis, historical-versus-current comparison, hazard indicator, and dashboard.
- **Final Project:** Integrated product, final report, live demonstration, and individual defense.